

Operations Research, Central Exercise 3

Revised Simplex & Sensitivity Analysis

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Revised Simplex

Matrix Formulation of an LP

$$\max c^T x \quad \text{s.t. } Ax = b, x \geq 0$$

- ▶ Partition columns of constraint matrix:

$$A = [B \ N]$$

- ▶ Partition variables and coefficients vectors:

$$x = \begin{pmatrix} x_B \\ x_N \end{pmatrix}, \quad c = \begin{pmatrix} c_B \\ c_N \end{pmatrix}$$

- ▶ System becomes:

$$Bx_B + Nx_N = b$$

Basis Transformation

- ▶ Multiply constraint by B^{-1} :

$$B^{-1}Bx_B + B^{-1}Nx_N = B^{-1}b$$

$$\Rightarrow x_B + B^{-1}Nx_N = B^{-1}b$$

- ▶ Define:

$$N' := B^{-1}N, \quad b' := B^{-1}b$$

- ▶ Then:

$$x_B = b' - N'x_N$$

Transformed Objective Function

- ▶ Objective:

$$z = c_B^T x_B + c_N^T x_N$$

- ▶ Substitute:

$$x_B = b' - N' x_N$$

- ▶ Then:

$$z = c_B^T b' + \left(c_N^T - c_B^T N' \right) x_N$$

- ▶ Rearranging:

$$z - \left(c_N^T - c_B^T N' \right) x_N = c_B^T b'$$

- ▶ Define:

$$c_N'^T := - \left(c_N - N'^T c_B \right)^T$$

- ▶ Final form:

$$z - c_N'^T x_N = c_B^T b'$$

Final Form

- ▶ Transformed system:

$$x_B = b' - N'x_N$$

- ▶ Objective in tableau form:

$$z - c_N'^T x_N = c_B^T b'$$

- ▶ Definitions:

$$b' = B^{-1}b, \quad N' = B^{-1}N$$

$$c_N'^T = - \left(c_N - N'^T c_B \right)^T$$

- ▶ Current basic feasible solution ($x_N = 0$):

$$x_B = b', \quad z = c_B^T b'$$

Comparison to Simplex Tableau

Basic Variable	Coefficient							Solution
	z	x_{B_1}	...	x_{B_m}	x_{N_1}	...	$x_{N_{n-m}}$	
z	1	0		0	$c_N'^T = -(c_N - N'^T c_B)^T$			$c_B^T B^{-1} b$
x_{B_1}		1			$N' = B^{-1} N$			$b' = B^{-1} b$
...			...					
x_{B_m}				1				

Important: Understand how each entry in the tableau is derived and what it represents (see, e.g., Exercise S3.3 from Retake 2025).

Feasibility and Optimality Conditions

- ▶ Transformed system:

$$x_B = b' - N'x_N, \quad z - c'_N{}^T x_N = c_B^T b'$$

- ▶ **Feasibility:**

$$x \geq 0 \iff b' = B^{-1}b \geq 0$$

- ▶ **Optimality (max problem):**

$$c'_N \geq 0 \quad \forall j \in N$$

- ▶ **Current solution:**

$$x = \begin{pmatrix} b' \\ 0 \end{pmatrix}, \quad z = c_B^T b'$$

- ▶ **Interpretation:**

- ▶ Feasible $\Rightarrow$ basic variables non-negative
- ▶ Optimal $\Rightarrow$ no improving non-basic direction

Revised Simplex Algorithm

1. Initialization:

- ▶ Find an initial basis B (basic variables)

2. Repeat:

- ▶ Compute B^{-1}
- ▶ Compute:

$$N' = B^{-1}N$$

- ▶ Compute reduced costs:

$$c'_N = -(c_N - N'^T c_B)$$

- ▶ **Entering variable:**

- ▶ Choose $j \in \arg \min_{k \in N} c'_k$ with $c'_j < 0$

- ▶ **Leaving variable:**

- ▶ Use N' , x_B and ratio test to maintain feasibility

- ▶ **Update:**

- ▶ Perform basis swap
- ▶ Update B , N , c_B , c_N

3. Until:

- ▶ $c'_N \geq 0$ (optimal solution), or
- ▶ problem is unbounded:

$$\exists j \in N \text{ with } c'_j < 0 \quad \text{and} \quad d := B^{-1}a_j \leq 0$$

Exercise D3.1: Revised Simplex

Problem:

$$\begin{aligned} \max \quad & 12x_1 + 20x_2 + 16x_3 + 12x_4 \\ & 4x_1 + 8x_2 + 2x_3 \leq 400 \\ \text{s.t.} \quad & 2x_1 + 5x_2 + 10x_3 + 4x_4 \leq 400 \\ & x_i \geq 0 \end{aligned}$$

Start with the feasible basis (x_1, x_6) .

Exercise D3.1: Revised Simplex

Problem:

$$\begin{aligned} \max \quad & 12x_1 + 20x_2 + 16x_3 + 12x_4 \\ & 4x_1 + 8x_2 + 2x_3 \leq 400 \\ \text{s.t.} \quad & 2x_1 + 5x_2 + 10x_3 + 4x_4 \leq 400 \\ & x_i \geq 0 \end{aligned}$$

Start with the feasible basis (x_1, x_6) .

Canonical Form:

$$\begin{aligned} \max \quad & 12x_1 + 20x_2 + 16x_3 + 12x_4 \\ & 4x_1 + 8x_2 + 2x_3 + x_5 = 400 \\ \text{s.t.} \quad & 2x_1 + 5x_2 + 10x_3 + 4x_4 + x_6 = 400 \\ & x_i \geq 0 \end{aligned}$$

$$A = \begin{pmatrix} 4 & 8 & 2 & 0 & 1 & 0 \\ 2 & 5 & 10 & 4 & 0 & 1 \end{pmatrix}, \quad b = \begin{pmatrix} 400 \\ 400 \end{pmatrix}$$

$$c = (12 \quad 20 \quad 16 \quad 12 \quad 0 \quad 0)^T$$

Iteration 1

$$A = \begin{pmatrix} 4 & 8 & 2 & 0 & 1 & 0 \\ 2 & 5 & 10 & 4 & 0 & 1 \end{pmatrix}$$

Basis:

$$x_B = (x_1, x_6)^T, \quad x_N = (x_2, x_3, x_4, x_5)^T$$

$$B = \begin{pmatrix} 4 & 0 \\ 2 & 1 \end{pmatrix}, \quad N = \begin{pmatrix} 8 & 2 & 0 & 1 \\ 5 & 10 & 4 & 0 \end{pmatrix}$$

Check: Valid basis? This is not the canonical basis $B = I$!

Iteration 1

$$A = \begin{pmatrix} 4 & 8 & 2 & 0 & 1 & 0 \\ 2 & 5 & 10 & 4 & 0 & 1 \end{pmatrix}$$

Basis:

$$x_B = (x_1, x_6)^T, \quad x_N = (x_2, x_3, x_4, x_5)^T$$

$$B = \begin{pmatrix} 4 & 0 \\ 2 & 1 \end{pmatrix}, \quad N = \begin{pmatrix} 8 & 2 & 0 & 1 \\ 5 & 10 & 4 & 0 \end{pmatrix}$$

Check: Valid basis? This is not the canonical basis $B = I$!

Yes, because $x_B = B^{-1}b \geq 0$.

Iteration 1: Computing B^{-1}

$$B = \begin{pmatrix} 4 & 0 \\ 2 & 1 \end{pmatrix}$$

$$[B \mid I] = \begin{pmatrix} 4 & 0 & 1 & 0 \\ 2 & 1 & 0 & 1 \end{pmatrix}$$

► $R_1 \leftarrow \frac{1}{4}R_1$

► $R_2 \leftarrow R_2 - 2R_1$

$$B^{-1} = \begin{pmatrix} \frac{1}{4} & 0 \\ -\frac{1}{2} & 1 \end{pmatrix}$$

Iteration 1: Computations

$$N' = B^{-1}N = \begin{pmatrix} 2 & \frac{1}{2} & 0 & \frac{1}{4} \\ 1 & 9 & 4 & -\frac{1}{2} \end{pmatrix}$$

$$c = (12 \quad 20 \quad 16 \quad 12 \quad 0 \quad 0)^T$$

$$c_B = \begin{pmatrix} 12 \\ 0 \end{pmatrix}, \quad c_N = (20 \quad 16 \quad 12 \quad 0)^T$$

$$c'_N = -(c_N - N'^T c_B) = \begin{pmatrix} 4 \\ -10 \\ -12 \\ 3 \end{pmatrix}$$

$$x_B = B^{-1}b = \begin{pmatrix} 100 \\ 200 \end{pmatrix}$$

Iteration 1: Pivot

- ▶ Reduced costs:

$$c'_N = (4, -10, -12, 3)^T$$

- ▶ **Entering variable:** Most negative reduced cost: $c'_4 = -12$.
Therefore: x_4 enters.
- ▶ Column of x_4 in N' :

$$N' = \begin{pmatrix} 2 & \frac{1}{2} & 0 & \frac{1}{4} \\ 1 & 9 & 4 & -\frac{1}{2} \end{pmatrix}, \quad N'_4 = \begin{pmatrix} 0 \\ 4 \end{pmatrix}$$

- ▶ Current basic solution:

$$x_B = \begin{pmatrix} x_1 \\ x_6 \end{pmatrix} = \begin{pmatrix} 100 \\ 200 \end{pmatrix}$$

- ▶ Ratio test: x_6 leaves
- ▶ **Pivot:** x_4 enters, x_6 leaves

Iteration 2

$$A = \begin{pmatrix} 4 & 8 & 2 & 0 & 1 & 0 \\ 2 & 5 & 10 & 4 & 0 & 1 \end{pmatrix}$$

New basis:

$$x_B = (x_1, x_4)^T, \quad x_N = (x_2, x_3, x_5, x_6)^T$$

$$B = \begin{pmatrix} 4 & 0 \\ 2 & 4 \end{pmatrix}, \quad N = \begin{pmatrix} 8 & 2 & 1 & 0 \\ 5 & 10 & 0 & 1 \end{pmatrix}$$

$$B^{-1} = \begin{pmatrix} \frac{1}{4} & 0 \\ -\frac{1}{8} & \frac{1}{4} \end{pmatrix}$$

Iteration 2: Computations

$$N' = B^{-1}N = \begin{pmatrix} 2 & \frac{1}{2} & \frac{1}{4} & 0 \\ \frac{1}{4} & \frac{9}{4} & -\frac{1}{8} & \frac{1}{4} \end{pmatrix}$$

$$c = (12 \quad 20 \quad 16 \quad 12 \quad 0 \quad 0)^T$$

$$c_B = \begin{pmatrix} 12 \\ 12 \end{pmatrix}, \quad c_N = (20 \quad 16 \quad 0 \quad 0)^T$$

$$c'_N = -(c_N - N'^T c_B) = \begin{pmatrix} 7 \\ 17 \\ \frac{3}{2} \\ 3 \end{pmatrix}$$

$$x_B = B^{-1}b = \begin{pmatrix} 100 \\ 50 \end{pmatrix}$$

Optimal Solution

- ▶ All reduced costs:

$$c'_N \geq 0 \quad (\text{no improving direction})$$

- ▶ Optimal basic solution:

$$x_1 = 100, \quad x_4 = 50$$

- ▶ Objective value:

$$Z = c_B^T B^{-1} b = 1800$$

Sensitivity Analysis

Sensitivity Analysis: Goal

- ▶ Study how the optimal solution changes when LP parameters change.
- ▶ Main questions:
 - ▶ What happens if the right-hand side b changes?
 - ▶ What happens if the objective coefficients c change?
 - ▶ How can we evaluate the value of resources?
 - ▶ How can we reoptimize efficiently after adding a new variable?
- ▶ Work with the current optimal basis B .

Recap: Simplex Tableau in Matrix Notation

Basic Variable	Coefficient							Solution
	z	x_{B_1}	...	x_{B_m}	x_{N_1}	...	$x_{N_{n-m}}$	
z	1	0		0	$c'_N = -(c_N - N^T c_B)^T$			$c_B^T B^{-1} b$
x_{B_1}		1			$N' = B^{-1} N$			$b' = B^{-1} b$
...			...					
x_{B_m}				1				

Current basic solution:

$$x_N = 0 \Rightarrow x_B = b' = B^{-1}b.$$

Feasibility: basic solution is non-negative

$$b' = B^{-1}b \geq 0.$$

Optimality for maximization: reduced costs are non-negative

$$c'_N = -(c_N - N'^T c_B) \geq 0$$

Change in the Right-Hand Side

Let the RHS change from b to

$$b + \Delta b.$$

The basis B stays fixed, so

$$x_B^{\text{new}} = B^{-1}(b + \Delta b) = b' + B^{-1}\Delta b.$$

- ▶ Reduced costs c'_N do not change if only b changes $\longleftrightarrow$ optimality not affected.
- ▶ Only feasibility may be lost.

Condition for feasibility:

$$b' + B^{-1}\Delta b \geq 0.$$

Shadow Prices

The objective value at the current basis is

$$z^* = c_B^T x_B = c_B^T B^{-1} b.$$

If b changes to $b + \Delta b$, then

$$z^{\text{new}} = z^* + c_B^T B^{-1} \Delta b.$$

Define shadow prices:

$$y^T := c_B^T B^{-1}.$$

y_i = marginal increase in z from one extra unit of resource i .

► y is determined by the current basis B .

Validity: Only valid while the current basis B remains optimal.

Shadow Prices and Reduced Costs

Given the shadow prices:

$$y^T = c_B^T B^{-1}$$

The **reduced costs** are defined as:

$$c'_j = - \left(c_j - a_j^T y \right), \quad j \in N.$$

Interpretation:

- ▶ c_j = direct profit of variable x_j
- ▶ $a_j^T y$ = value of resources used (at shadow prices)

Economic meaning:

$$c_j - a_j^T y = \text{net profit of introducing } x_j.$$

- ▶ If $c'_j \geq 0$: not profitable $\Rightarrow$ keeping $x_j = 0$ is optimal
- ▶ If $c'_j < 0$: profitable $\Rightarrow$ enter basis

Shadow Prices in the Simplex Tableau

Reduced costs:

$$c'_j = -(c_j - a_j^T y), \quad \text{with } y^T = c_B^T B^{-1}$$

For slack variables s_i :

$$c_{s_i} = 0, \quad a_{s_i} = e_i \Rightarrow c'_{s_i} = y_i$$

- Entries under slack variables in the objective row = shadow prices

Case	Slack s_i	Constraint	Shadow Price y_i
Non-binding	$s_i > 0$	Not tight (unused resource)	$y_i = 0$
Binding	$s_i = 0$	Tight (fully used resource)	$y_i > 0$

Complementary slackness condition (next lecture):

$$y_i \cdot s_i = 0$$

Change in the Objective Coefficients

Now keep A , b fixed, but change the objective coefficients c .

If the basis remains the same, then

$$x_B = B^{-1}b$$

does not change.

But the reduced costs do change:

$$c'_N = -(c_N - N'^T c_B).$$

If c_B or c_N changes by Δ_d , recompute c'_N , i.e., $c'_N(\Delta_d)$.

Condition for the current basis to remain optimal:

$$c'_N(\Delta_d) \geq 0$$

Reoptimization After Adding a New Variable

Suppose a new variable x_{new} with column a_{new} and objective coefficient c_{new} is added.

Do not solve the LP from scratch!

Instead: Use the current optimal basis B and compute:

$$\bar{a} = B^{-1}a_{new}.$$

The reduced cost of the new variable is

$$c'_{new} = -(c_{new} - \bar{a}^T c_B).$$

- ▶ If $c'_{new} \geq 0$, the current basis remains optimal.
- ▶ If $c'_{new} < 0$, the new variable should enter the basis.

Adding a new constraint is harder because it may destroy feasibility!

Exercise D3.2, a): Waldgeist Distillery

Resources	Himmelrot	Nachtblau	Capacity
Raspberry	2	1	20
Blueberry	2	3	18
Strawberry	1	1/2	8
Unit price	15	20	
Unit cost	10	16	

$$\begin{aligned} \max & 5x_1 + 4x_2 \\ \text{s.t. } & 2x_1 + x_2 \leq 20 \\ & 2x_1 + 3x_2 \leq 18 \\ & x_1 + 1/2x_2 \leq 8 \\ & x \in \mathbb{R}_{\geq 0}^2 \end{aligned}$$

Exercise D3.2, a): Waldgeist Distillery

Decision variables:

x_1 = Himmelrot bottles produced, x_2 = Nachtblau bottles produced

Maximize profit:

$$15x_1 + 20x_2 - (10x_1 + 16x_2) = 5x_1 + 4x_2$$

$$\max 5x_1 + 4x_2$$

Capacity constraints:

$$2x_1 + x_2 \leq 20 \quad (\text{raspberries})$$

$$2x_1 + 3x_2 \leq 18 \quad (\text{blueberries})$$

$$x_1 + \frac{1}{2}x_2 \leq 8 \quad (\text{strawberries})$$

Non-negativity constraint:

$$x_1, x_2 \geq 0$$

b) Final Simplex Tableau

	x_1	x_2	x_3	x_4	x_5	Res.
Z			0	0.75	3.5	
			1	0	-2	
x_2			0	0.5	-1	
x_1			0	-0.25	1.5	7.5

b) Final Simplex Tableau

	x_1	x_2	x_3	x_4	x_5	Res.
Z	0	0	0	0.75	3.5	41.5
x_3	0	0	1	0	-2	4
x_2	0	1	0	0.5	-1	1
x_1	1	0	0	-0.25	1.5	7.5

What is the economic interpretation of x_3 , x_4 , x_5 ?

b) Final Simplex Tableau

	x_1	x_2	x_3	x_4	x_5	Res.
Z	0	0	0	0.75	3.5	41.5
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x_2	0	1	0	0.5	-1	1
x_1	1	0	0	-0.25	1.5	7.5

What is the economic interpretation of x_3, x_4, x_5 ?

- ▶ x_3 : unused kg of raspberries
- ▶ x_4 : unused kg of blueberries
- ▶ x_5 : unused kg of strawberries

c) Shadow Price of Raspberries

What is the maximum amount that Steffen should pay for 1kg of additional raspberries? Justify your answer.

	x_1	x_2	x_3	x_4	x_5	Res.
Z	0	0	0	0.75	3.5	41.5
x_3	0	0	1	0	-2	4
x_2	0	1	0	0.5	-1	1
x_1	1	0	0	-0.25	1.5	7.5

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	x_1	x_2	x_3	x_4	x_5	Res.
Z	0	0	0	0.75	3.5	41.5
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x_2	0	1	0	0.5	-1	1
x_1	1	0	0	-0.25	1.5	7.5

The shadow price of an additional unit of raspberries is 0. (There are still 4 units of raspberries left.) Therefore, Steffen should not spend anything on additional raspberries.

d) Selling Price Limits for Nachtblau - Analyze Change in Objective Coefficients

Calculate the limits for the **selling price** of Nachtblau within which the optimal basis does not change. Explain in detail how the optimal production schedule and resource stocks change, if the selling price violates these limits.

Recap: Objective Coefficient Sensitivity

Change one objective coefficient:

$$c_j \rightarrow c_j + \Delta$$

- ▶ A, b stay fixed $\Rightarrow$ feasibility does not change.
- ▶ Only reduced costs change.
- ▶ Same basis remains optimal iff:

$$c'_N(\Delta) \geq 0$$

If the changed variable is basic:

$$c_B \rightarrow c_B + \Delta e_i$$

$$c'_N(\Delta) = c'_N + \Delta(N'^T e_i)$$

Tableau rule: Modify the Z -row, eliminate the basic variable again, then require all reduced costs ≥ 0 .

d) Selling Price Limits for Nachtblau - Analyze Change in Objective Coefficients

Let the price of Nachtblau change by Δ .

	x_1	x_2	x_3	x_4	x_5	<i>RHS</i>
Z	0	$-\Delta$	0	0.75	3.5	41.5
x_3	0	0	1	0	-2	4
x_2	0	1	0	0.5	-1	1
x_1	1	0	0	-0.25	1.5	7.5

d) Selling Price Limits for Nachtblau - Analyze Change in Objective Coefficients

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	x_1	x_2	x_3	x_4	x_5	<i>RHS</i>
Z	0	$-\Delta$	0	0.75	3.5	41.5
x_3	0	0	1	0	-2	4
x_2	0	1	0	0.5	-1	1
x_1	1	0	0	-0.25	1.5	7.5

Transform such that the entry of x_2 in Z row is 0 (x_2 is basic):

	x_1	x_2	x_3	x_4	x_5	<i>RHS</i>
Z	0	0	0	$0.75 + \frac{\Delta}{2}$	$3.5 - \Delta$	$41.5 + \Delta$

d) Selling Price Limits for Nachtblau - Analyze Change in Objective Coefficients

Let the price of Nachtblau change by Δ .

	x_1	x_2	x_3	x_4	x_5	<i>RHS</i>
Z	0	$-\Delta$	0	0.75	3.5	41.5
x_3	0	0	1	0	-2	4
x_2	0	1	0	0.5	-1	1
x_1	1	0	0	-0.25	1.5	7.5

Transform such that the entry of x_2 in Z row is 0 (x_2 is basic):

	x_1	x_2	x_3	x_4	x_5	<i>RHS</i>
Z	0	0	0	$0.75 + \frac{\Delta}{2}$	$3.5 - \Delta$	$41.5 + \Delta$

For the current basis to remain optimal, it must hold $c'_N \geq 0$, i.e.,

$$0.75 + \frac{\Delta}{2} \geq 0 \text{ and } 3.5 - \Delta \geq 0 \iff -1.5 \leq \Delta \leq 3.5$$

d) Selling Price Limits for Nachtblau - Analyze Change in Objective Coefficients

Let the price of Nachtblau change by Δ .

	x_1	x_2	x_3	x_4	x_5	<i>RHS</i>
Z	0	$-\Delta$	0	0.75	3.5	41.5
x_3	0	0	1	0	-2	4
x_2	0	1	0	0.5	-1	1
x_1	1	0	0	-0.25	1.5	7.5

Transform such that the entry of x_2 in Z row is 0 (x_2 is basic):

	x_1	x_2	x_3	x_4	x_5	<i>RHS</i>
Z	0	0	0	$0.75 + \frac{\Delta}{2}$	$3.5 - \Delta$	$41.5 + \Delta$

For the current basis to remain optimal, it must hold $c'_N \geq 0$, i.e.,

$$0.75 + \frac{\Delta}{2} \geq 0 \text{ and } 3.5 - \Delta \geq 0 \iff -1.5 \leq \Delta \leq 3.5$$

$$p_N \in [20 - 1.5, 20 + 3.5] = [18.5, 23.5]$$

d) Selling Price Limits for Nachtblau

Price can vary within $p_N \in [18.5, 23.5]$ for $\Delta \in [-1.5, 3.5]$ without basis change.

If the price of Nachtblau **falls below** 18.5, i.e., if $\Delta < -1.5$, then:

$$0.75 + \frac{\Delta}{2} < 0$$

	x_1	x_2	x_3	x_4	x_5	<i>RHS</i>
Z	0	0	0	$0.75 + \frac{\Delta}{2}$	$3.5 - \Delta$	$41.5 + \Delta$
x_3	0	0	1	0	-2	4
x_2	0	1	0	0.5	-1	1
x_1	1	0	0	-0.25	1.5	7.5

So the current basis is no longer optimal.

Basis change: x_4 enters, x_2 leaves

New optimal interpretation: produce 8 bottles Himmelrot, 0 Nachtblau; 4 raspberries, 2 blueberries, 0 strawberries left.

d) Selling Price Limits for Nachtblau

Price can vary within $p_N \in [18.5, 23.5]$ for $\Delta \in [-1.5, 3.5]$ without basis change.

If the price of Nachtblau **falls above** 23.5, i.e., if $\Delta > 3.5$, then:

$$3.5 - \Delta < 0$$

	x_1	x_2	x_3	x_4	x_5	<i>RHS</i>
Z	0	0	0	$0.75 + \frac{\Delta}{2}$	$3.5 - \Delta$	$41.5 + \Delta$
x_3	0	0	1	0	-2	4
x_2	0	1	0	0.5	-1	1
x_1	1	0	0	-0.25	1.5	7.5

So the current basis is no longer optimal.

Basis change: x_5 enters, x_1 leaves

New optimal interpretation: produce 0 bottles Himmelrot, 6 Nachtblau; 14 raspberries, 0 blueberries, 5 strawberries left.

e) Supply of blueberries/strawberries varies - Analyze Change in RHS

Within what limits can the blueberry/strawberry **capacity** move so that both products can continue to be produced?

Recap: RHS Sensitivity Analysis

Change in RHS:

$$b \rightarrow b + \Delta b$$

Updated basic solution:

$$x_B^{new} = B^{-1}(b + \Delta b) = b' + B^{-1}\Delta b$$

- ▶ Reduced costs c'_N do **not** change
- ▶ Only feasibility may be lost

For a single constraint i :

- ▶ Change RHS by Δ_i
- ▶ Then:

$$x_B^{new} = b' + \Delta_i \cdot (B^{-1}e_i)$$

- ▶ In the tableau: use the column of the corresponding slack variable x_i

Condition for same basis:

$$x_B^{new} \geq 0$$

e) (i) Blueberry Capacity Allowed Variation - **SKIP**

Let Δ_B change the RHS of the blueberry constraint.

Using the final tableau:

	x_1	x_2	x_3	x_4	x_5	<i>RHS</i>
Z	0	0	0	0.75	3.5	41.5
x_3	0	0	1	0	-2	4
x_2	0	1	0	0.5	-1	1
x_1	1	0	0	-0.25	1.5	7.5

Updated solution becomes $x_2 = 1 + 0.5\Delta_B$ and $x_1 = 7.5 - 0.25\Delta_B$

Both products produced iff

$$x_2 = 1 + 0.5\Delta_B > 0, \quad x_1 = 7.5 - 0.25\Delta_B > 0$$

Thus:

$$-2 < \Delta_B < 30$$

Since original blueberry capacity is 18:

$$B_{\text{cap}} \in (18 - 2, 18 + 30) = (16, 48)$$

e) (ii) Strawberry Capacity Allowed Variation

Let Δ_S change the RHS of the strawberry constraint.
Using the final tableau:

	x_1	x_2	x_3	x_4	x_5	RHS
Z	0	0	0	0.75	3.5	41.5
x_3	0	0	1	0	-2	4
x_2	0	1	0	0.5	-1	1
x_1	1	0	0	-0.25	1.5	7.5

Updated solution becomes

$$x_3 = 4 - 2\Delta_S, \quad x_2 = 1 - \Delta_S, \quad x_1 = 7.5 + 1.5\Delta_S$$

Feasibility and producing both products require:

$$4 - 2\Delta_S \geq 0, \quad 1 - \Delta_S > 0, \quad 7.5 + 1.5\Delta_S > 0$$

Thus:

$$-5 < \Delta_S < 1$$

$$S_{\text{cap}} \in (8 - 5, 8 + 1) = (3, 9)$$

e) (iii) Strawberry Capacity Too Low ($\Delta_S = -5$)

Step 1: Check feasibility

$$x_1 = 7.5 + 1.5(-5) = 0 \quad \Rightarrow \quad x_1 \text{ leaves the basis}$$

Step 2: Restore optimality

Let t be the new coefficient of x_1 in the Z -row:

	x_1	x_2	x_3	x_4	x_5
$Z(t)$	t	0	0	$0.75 - 0.25t$	$3.5 + 1.5t$
x_1	1	0	0	-0.25	1.5

Step 3: Optimality condition: $0.75 - 0.25t \geq 0 \quad \Rightarrow \quad t \leq 3$

Step 4: Pivot At $t = 3$: x_4 enters, x_1 leaves

Result (new shadow prices / reduced costs): $y = (0, 0, 8)$

$\Rightarrow$ Strawberry shadow price increases from 3.5 to 8

e) (iii) Strawberry Capacity Too High ($\Delta_5 = 1$) - **SKIP**

Step 1: Check feasibility

$$x_2 = 1 - 1 = 0 \Rightarrow x_2 \text{ leaves the basis}$$

Step 2: Restore optimality

Let t be the new coefficient of x_2 in the Z -row:

	x_1	x_2	x_3	x_4	x_5
$Z(t)$	0	t	0	$0.75 + 0.5t$	$3.5 - t$
x_2	0	1	0	0.5	-1

Step 3: Optimality condition: $3.5 - t \geq 0 \Rightarrow t \leq 3.5$

Step 4: Pivot At $t = 3.5$: x_5 enters, x_2 leaves

Result (new shadow prices / reduced costs): $y = (0, 2.5, 0)$

$\Rightarrow$ Blueberry shadow price increases from 0.75 to 2.5

f) Berry Exchange Optimization

Exchange Δ units between blueberries and strawberries.

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Exchange Δ units between blueberries and strawberries.

From the final tableau: $Z = 41.5 + 0.75\Delta - 3.5\Delta = 41.5 - 2.75\Delta$

	x_1	x_2	x_3	x_4	x_5	Res.
Z	0	0	0	0.75	3.5	41.5
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Feasibility:

$$x_3 = 4 + 2\Delta \geq 0, \quad x_2 = 1 + 1.5\Delta \geq 0, \quad x_1 = 7.5 - 1.75\Delta \geq 0$$

Thus:

$$\Delta \in \left[-\frac{2}{3}, \frac{30}{7} \right]$$

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Feasibility:

$$x_3 = 4 + 2\Delta \geq 0, \quad x_2 = 1 + 1.5\Delta \geq 0, \quad x_1 = 7.5 - 1.75\Delta \geq 0$$

Thus:

$$\Delta \in \left[-\frac{2}{3}, \frac{30}{7} \right]$$

Since Z decreases in Δ , choose:

$$\Delta = -\frac{2}{3} \implies Z = 41.5 - 2.75 \left(-\frac{2}{3} \right) = 43.33$$

g) New Product: Fruchttraum I

Fruchttraum uses:

4 raspberries, 2 blueberries, 3 strawberries

Production cost:

13

Current shadow prices:

$(0, 0.75, 3.5)$

g) New Product: Fruchttraum I

Fruchttraum uses:

4 raspberries, 2 blueberries, 3 strawberries

Production cost:

13

Current shadow prices:

$(0, 0.75, 3.5)$

Check validity of shadow prices:

- ▶ Raspberries: at most 4 available (see optimal tableau) $\Rightarrow$ OK
- ▶ Blueberries: valid for decrease ≤ 2 (see e(i)) $\Rightarrow$ OK
- ▶ Strawberries: valid for decrease ≤ 5 (see e(ii)) $\Rightarrow$ OK

g) New Product: Fruchttraum I

Fruchttraum uses:

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Production cost:

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Current shadow prices:

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Check validity of shadow prices:

- ▶ Raspberries: at most 4 available (see optimal tableau) $\Rightarrow$ OK
- ▶ Blueberries: valid for decrease ≤ 2 (see e(i)) $\Rightarrow$ OK
- ▶ Strawberries: valid for decrease ≤ 5 (see e(ii)) $\Rightarrow$ OK

Resource value (opportunity cost):

$$a_{new}^T y = 4 \cdot 0 + 2 \cdot 0.75 + 3 \cdot 3.5 = 12$$

Minimum price:

$$12 + 13 = 25$$

g) New Product: Fruchttraum II

Now Fruchttraum uses:

4 raspberries, 2 blueberries, 6 strawberries

Current strawberry shadow price is valid only for a decrease of 5 units.

For first 5 strawberry units use current shadow prices (0, 0.75, 3.5):

$$4 \cdot 0 + 2 \cdot 0.75 + 5 \cdot 3.5 = 19$$

For the extra 1 strawberry unit, use the new shadow price (e(iii)):

$$1 \cdot 8 = 8$$

Total resource value:

$$19 + 8 = 27$$

Add production cost:

$$27 + 13 = 40 \implies \boxed{\text{Minimum price} = 40}$$

h) Integrality Constraints

If bottles must be produced in integer quantities, add:

$$x_1, x_2 \in \mathbb{Z}_{\geq 0}$$

Then the model becomes an **integer program**.

The LP optimum is:

$$Z_{LP}^* = 41.5$$

Next IP Modeling Lectures: Since every integer feasible solution is also feasible for the LP relaxation:

$$Z_{IP}^* \leq Z_{LP}^*$$

41.5 is an upper bound for the integer program

Notes

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